Kernel trick

- $\alpha_i \neq 0$ *only* for points in the gutter, also known as support vectors
- Decision for $\boldsymbol{x_j}$: $\boldsymbol{w x_j} + b = \sum \alpha_i y_i \boldsymbol{x_i x_j} + b$
- We can change the representation of $\boldsymbol{x_i}$ to be $\phi(\boldsymbol{x_i})$
- We do not need direct access to $\phi(\boldsymbol{x_i})$
  - Instead we define $k(\boldsymbol{x_i}, \boldsymbol{x_j}) = \phi(\boldsymbol{x_i})\phi(\boldsymbol{x_j})$
  - The function *k* is called kernel
- $f(\boldsymbol{x}) = \sum \alpha_i y_i k(\boldsymbol{x_i}, \boldsymbol{x}) + b$
  - This function is nonlinear with respect to $\boldsymbol{x}$

Advantages of the kernel trick

- The kernel trick is powerful for two reasons.
  - it allows us to learn models that are nonlinear as a function of $\boldsymbol{x}$ using convex optimization techniques that are guaranteed to converge efficiently
  - the kernel function often admits an implementation that is significantly more computational efficient than naively constructing two $\phi(\boldsymbol{x})$ vectors and explicitly taking their dot product.
    - In some cases, $\phi(\boldsymbol{x})$ can even be infinite dimensional

Gaussian kernel

- Gaussian kernel $k(\boldsymbol{u}, \boldsymbol{v}) = \mathcal{N}(\boldsymbol{u} - \boldsymbol{v}; 0, \sigma^2 \boldsymbol{I})$
  - Aka RBF: Radial Basis Function

- We can think of the Gaussian kernel as performing a kind of template matching.
  - A training example x associated with training label y becomes a template for class y.
  - When a test point x' is near x according to Euclidean distance, the Gaussian kernel has a large response, indicating that x' is very similar to the x template.
  - The model then puts a large weight on the associated training label y.
- Overall, the prediction will combine many such training labels weighted by the similarity of the corresponding training examples.

SVM limitations

- high computational cost of training when the dataset is large.
- The current deep learning renaissance began when Hinton et al. (2006) demonstrated that a neural network could outperform the RBF kernel SVM on the MNIST benchmark.

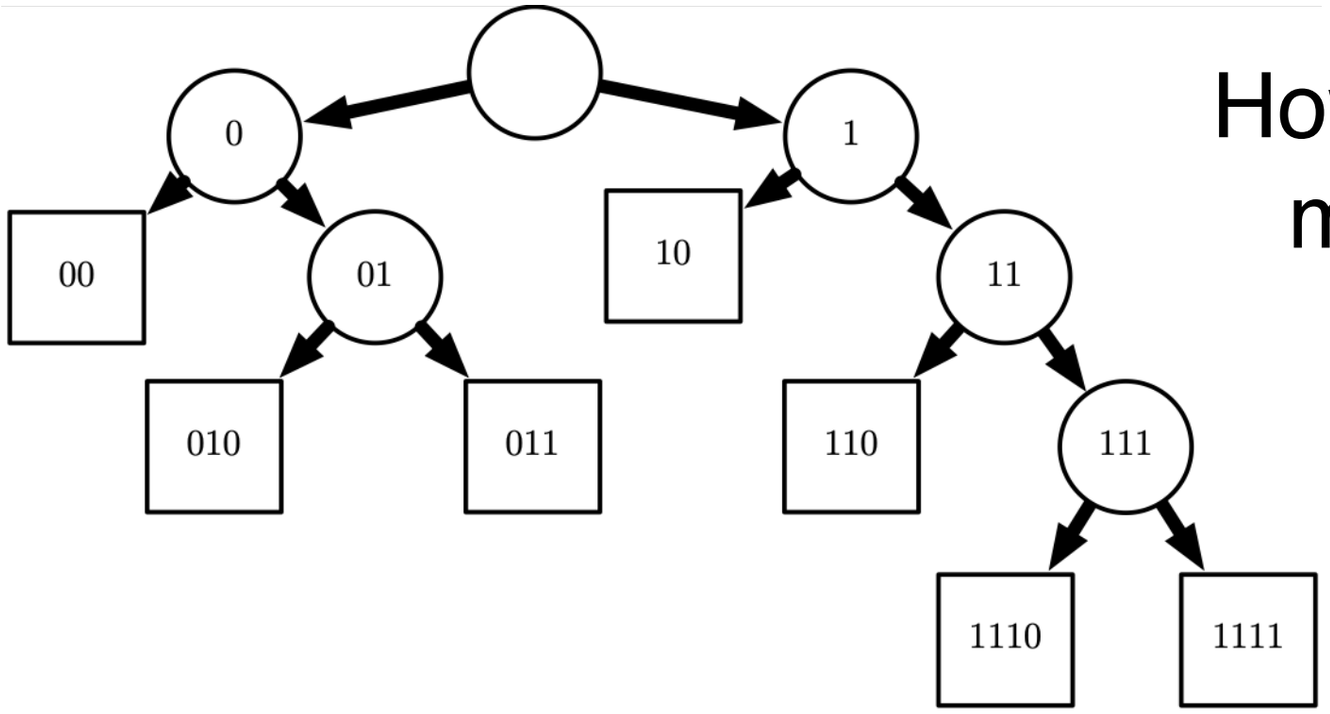
K-Nearest Neighbors

- There is not even really a training stage or learning process.
  - We find the k-nearest neighbors to $\boldsymbol{x}$ in the training data $\boldsymbol{X}$. We then return the average of the corresponding y values in the training set
- In the case of classification, we can average over one-hot code vectors
  - We can then interpret the average over these one-hot codes as giving a probability distribution over classes

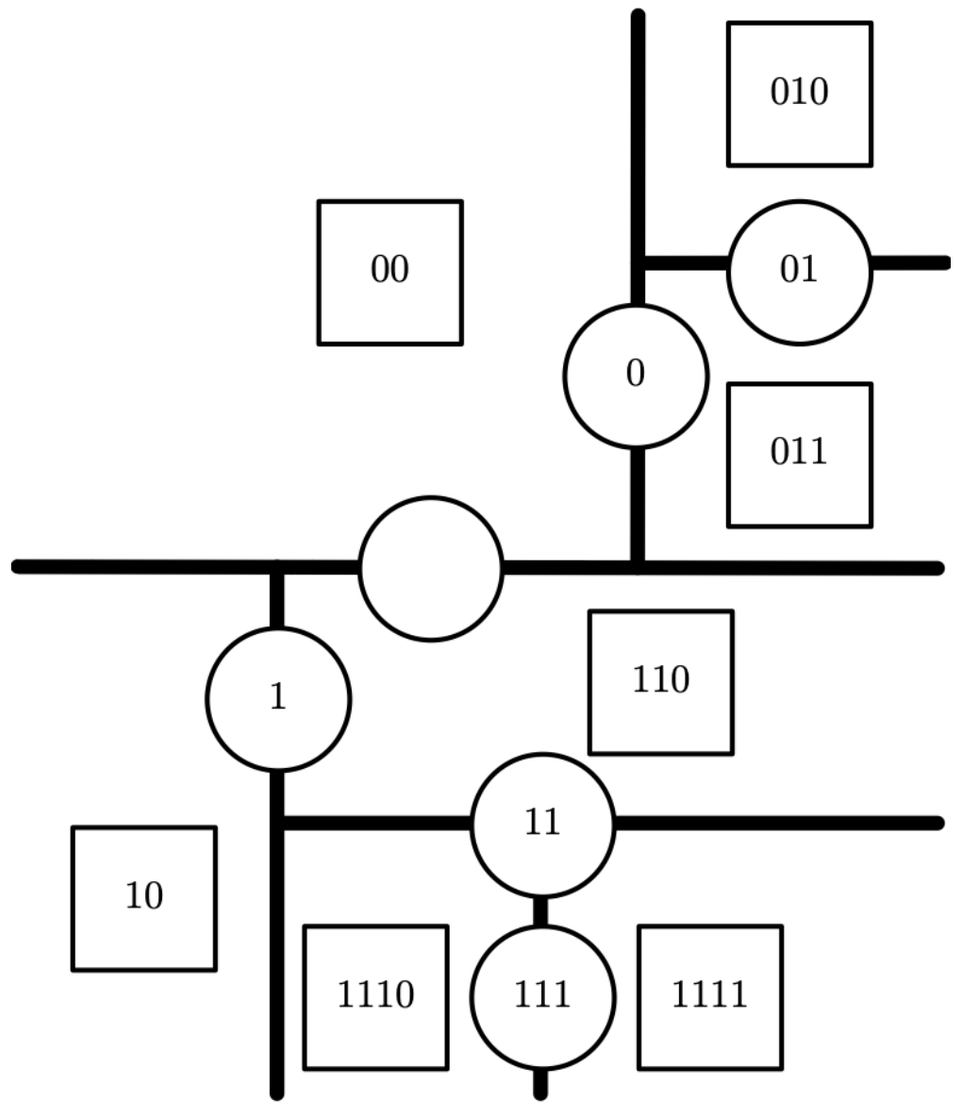
K-NN limitations

- It cannot learn that one feature is more discriminative than another.
  - For example, imagine we have a regression task with $x \in \mathbb{R}^{100}$ drawn from an isotropic Gaussian distribution,
  - but only a single variable is relevant to the output $y = x_1$
  - Nearest neighbor regression will not be able to detect this simple pattern.

Decision Trees



How a decision tree might divide $\mathbb{R}^2$.



Each leaf requires at least one training example to define,
so it is not possible for the decision tree to learn a function that has more local maxima than the number of training examples.

Figure 5.7

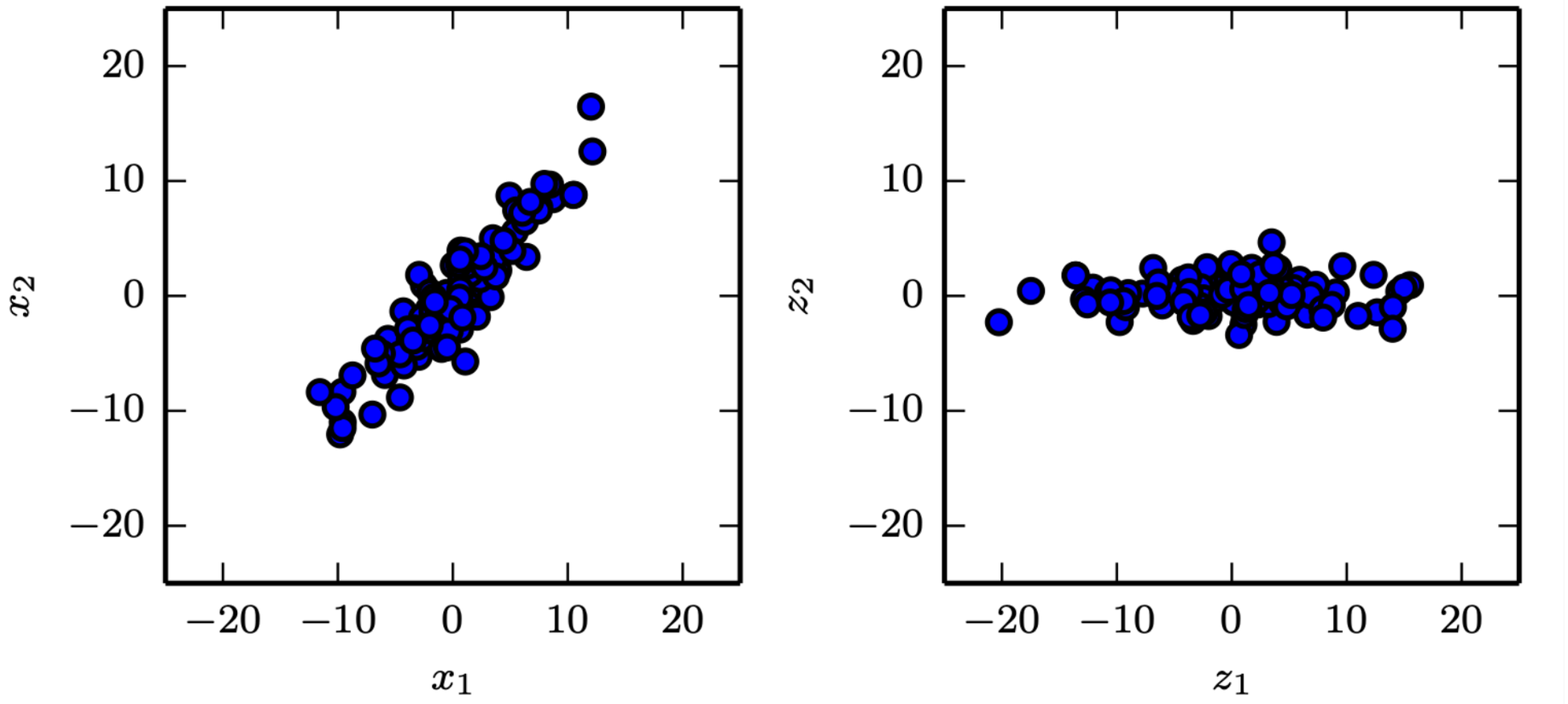
Decision trees limitations

- It struggles to solve some problems that are easy even for logistic regression.
- If we have a two-class problem and the positive class occurs wherever $x_2 > x_1$
  - the decision boundary is not axis-aligned. The decision tree will thus need to approximate the decision boundary with many splits, implementing a step function that constantly walks back and forth across the true decision function.

Unsupervided learning

- Density estimation
- Learning to draw samples from a distribution
- Learning to denoise data from some distribution
- Clustering the data into groups of related examples
- Simpler representation:
  - Lower-dimensional representation
  - Sparse representation
  - Independent representation

Principal Components Analysis



- lower dimensionality
- **no linear correlation**